

1 Physical Primitives of Mind: Modeling
2 Consciousness via Symmetries and Landauer
3 Erasure

4 Sebastian Bobadilla-Suarez
`sebastian.bobadilla.s@gmail.com`

5 31 August 2026

6 **Abstract**

7 We present a machine-checked synthesis of established physical
8 results into a single conditional route from the finiteness of a local-
9 ized system’s phase space to a unified cognitive fixed point. No new
10 fundamental physics is proposed: Kibble’s classification of topolog-
11 ical defects, Landauer’s erasure bound, the bound of Still and col-
12 leagues relating dissipation to nonpredictive information, the thresh-
13 old $K_c = 2D$ for the noisy mean-field Kuramoto model, and Banach’s
14 fixed-point theorem are all imported. What is new is that they are
15 stated in one formal language, their mutual consistency mechanically
16 checked, and the points at which the route relies on stipulation rather

17 than derivation located exactly. One conditional theorem in Lean 4
 18 composes eight named hypotheses to its endpoint; the unproved steps
 19 are its arguments rather than sentences about it, and they are jointly
 20 satisfiable rather than jointly empty. Two links are marked rather
 21 than argued around: the identification of the posited continuum ker-
 22 nel with the endogenous electromagnetic field of cortex is an empirical
 23 commitment with four stated failure conditions, and the identification
 24 of a global section of the probability sheaf with the unity of experience
 25 is a stipulation. The framework is falsifiable in one place that is its
 26 own. Recovery from sudden awakening should follow a delayed sig-
 27 moid with a square-root cusp rather than an exponential, because the
 28 slow variable gating recovery is astrocytic volume rather than cortical
 29 phase realignment, and throughout that recovery the concentration
 30 and the coherence of the phase distribution must collapse onto the
 31 parameter-free curve $r = I_1(a)/I_0(a)$.

32 **1 Introduction**

33 The pursuit of a physical theory of consciousness typically involves search-
 34 ing for the biological correlates of mind or treating subjective experience
 35 as an abstract computational algorithm. This paper asks a more funda-
 36 mental question: if we assume only basic physical primitives—continuous
 37 symmetries, geometry, and thermodynamics—is the emergence of a unified,
 38 perceiving entity a natural physical attractor?

39 We propose a framework in which the structural properties of conscious-
 40 ness are organized as candidate physical attractor states, running from con-
 41 tinuous symmetries and boundaries, through dissipation and physical reso-
 42 nance, to the macroscopic electromagnetic unity proposed by field theories
 43 of mind. This is not a derivation from basic principles: the formal result is
 44 a conditional composition schema whose eight unproved arrows are explicit
 45 assumptions. To avoid critiques of “math theatre,” we delineate between lit-
 46 eral physical mechanisms (thermodynamic erasure via Landauer’s principle,
 47 boundaries from spontaneous symmetry breaking) and theoretical mappings,
 48 using Lean 4 formalization as our mathematical anchor. We adopt a perspec-
 49 tive aligned with Russellian monism: objective measurements of macroscopic
 50 electromagnetic fields constitute the third-person “outside,” while the sys-
 51 tem’s internal physical deformation to reach structural resonance constitutes
 52 the first-person intrinsic experience.

53 **What is new here, and what is not**

54 A framework paper about consciousness invites a particular suspicion, and it
 55 is better answered at the start than at the end. **No new physics is derived**
 56 **here.** Every physical result the chain invokes is established, and most of
 57 it is old: Picard–Lindelöf for the well-posedness of the coupled dynamics
 58 (Coddington & Levinson, 1955); Kibble’s classification of topological defects
 59 by the homotopy of the vacuum manifold (Kibble, 1976); Landauer’s bound
 60 on the heat cost of erasure (Landauer, 1961); Gibbs’ inequality for the relative

61 entropy (Cover & Thomas, 2006); Barbalat’s lemma for the asymptotics of a
 62 dissipating trajectory (Barbalat, 1959); the threshold $K_c = 2D$ for the noisy
 63 mean-field Kuramoto model (Sakaguchi, 1988); the bound of Still, Sivak,
 64 Bell and Crooks relating dissipation to nonpredictive information (Still et
 65 al., 2012); the identity of Kawai, Parrondo and Van den Broeck equating
 66 dissipated work with the relative entropy between forward and time-reversed
 67 ensembles (Kawai et al., 2007); and Banach’s fixed-point theorem (Banach,
 68 1922). A physicist reading this paper will recognize all of it. Conceding the
 69 point costs nothing.

70 What is new is of a different kind, and it is a smaller claim. These
 71 results are stated in one formal language, their mutual consistency mechan-
 72 ically checked, and the places where the chain relies on stipulation rather
 73 than derivation located exactly. The development declares **no axioms**:
 74 `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`.
 75 Every physical postulate is a field of a class each model discharges for it-
 76 self, and every class is inhabited by an explicit witness, so no theorem
 77 is vacuous. The links form a single conditional composition theorem —
 78 `chain` in `Chain.lean` — with **eight named hypotheses**: four formaliza-
 79 tion gaps, two modelling assumptions, two physical commitments. Its con-
 80 clusion `UnifiedSelf` says the sheaf-glued section is the unique fixed point
 81 of self-prediction, rather than `Unity` and `Self` being unrelated. The theo-
 82 rem `chain_hypotheses_jointly_satisfiable` discharges all eight on one
 83 toy substrate, ruling out inconsistency without evidence that cortex realizes

84 the composition.

85 The schema exposes two missing arrows: the capacity bound implies the
86 heat cost directly, skipping the boundary, and neither of the two theorem-
87 arrows is one the framework’s usual narration draws.

88 The framework is not thereby unfalsifiable, and one of its predictions is
89 its own rather than a rival’s. Section 12 argues that recovery from sudden
90 awakening should follow a *delayed sigmoid* rather than an exponential, be-
91 cause the variable gating recovery is astrocytic volume and cerebrospinal-fluid
92 clearance, on a timescale of seconds to minutes, while the cortical rhythms
93 whose realignment is usually taken to constitute the recovery move in mil-
94 liseconds. That is a claim about functional form; it is fittable against data
95 of a kind already collected; and a purely electrical account of the same re-
96 covery predicts a different form. Because a delayed sigmoid is what any slow
97 saturating gate would give, that section also states the sharper claim the
98 same mathematics supports: at every instant of the recovery the concentra-
99 tion and the coherence of the phase distribution must lie on the fixed curve
100 $r = I_1(a)/I_0(a)$, the same curve for every subject, with no free parameters
101 and with the threshold read off the ratio a/r rather than fitted.

102 2 The chain, and the status of each of its links

103 The machine-checked dependency accounting is carried out in Lean 4, build-
104 ing on recent efforts to formalize consciousness and brain semantics such as

105 the ‘OnlyOne’ system (Scherf, 2026) and sheaf-theoretic models of neural
 106 manifolds (Inoué, 2026). Figure 1 shows the schema with the status of every
 107 node *and of every arrow* marked, and Table 1 states in seven rows what
 108 each link assumes. The complete claim-by-claim map to Lean identifiers is
 109 Table S1 in the supplement; the main text does not depend on reading it.

110 One result about the formalization itself is worth stating here rather than
 111 in the supplement, because it is what makes the accounting mean anything.
 112 An axiom that is inconsistent silently makes every theorem downstream of
 113 it vacuous while leaving the build green, and the failure has one shape: *an*
 114 *axiom that constrains a symbol it does not itself bind*. Such a declaration
 115 is not a postulate about one system but a constraint on every model of the
 116 class the symbol belongs to, including models built to violate it, and a class
 117 that has such a model is thereby inconsistent. Three tempting global axiom
 118 shapes — pointwise vacuum minimization with the energy functional as an
 119 implicit argument, a Landauer heat identity equating a free class field with a
 120 bath-entropy difference for every instance, and an entropy-production bound
 121 asserted against all pairs of distributions — each prove **False**, from that one
 122 root cause. The rule that prevents it is the rule we follow throughout: *a*
 123 *physical postulate that mentions a class field must be a field of that class,*
 124 *never a standalone axiom quantified over all instances*. A postulate is then
 125 an obligation each model discharges, not a global claim about every model.
 126 The supplement gives the three refutations and the corresponding witness
 127 constructions, which are what keep the classes inhabited and their theorems

128 non-vacuous.

129 3 Symmetries as setting, and the capacity of 130 a finite phase space

131 Our starting premise relies on fundamental geometry. Continuous spacetime
132 symmetries act upon field states, and the invariance of the system's total
133 energy under that action is recorded by the class `SymmetryInvariantAction`
134 (`Phase1Primitives.lean`). Physically, it is such an invariance that Noether's
135 theorem converts into the conservation of energy and momentum. The scope
136 of the *formal* claim is narrower and we state it plainly: the invariance is a
137 class field, an assumption a model must discharge, and no conserved quantity
138 is ever constructed from it. There is no Noether theorem in our Lean, and
139 nothing downstream depends on one. The class is inhabited by a double-
140 well potential $V(v) = (v^2 - 1)^2$ on a two-point spacetime with the $\mathbb{Z}_2$ action
141 $v \mapsto \pm v$, whose vacuum manifold $\{-1, 1\}$ is proved degenerate with the two
142 minima exchanged by the symmetry. That is symmetry breaking exhibited
143 rather than assumed; it is still not a conserved quantity. Symmetries and the
144 metric are therefore *setting* rather than content, and Figure 1 marks them
145 as such.

146 The other half of the premise is not setting. Because phase space is finite,
147 any localized physical system possesses a mathematical limit on the informa-
148 tion it can embody at any moment, and three theorems in `Phase1PhaseSpaceCapacity.lean`

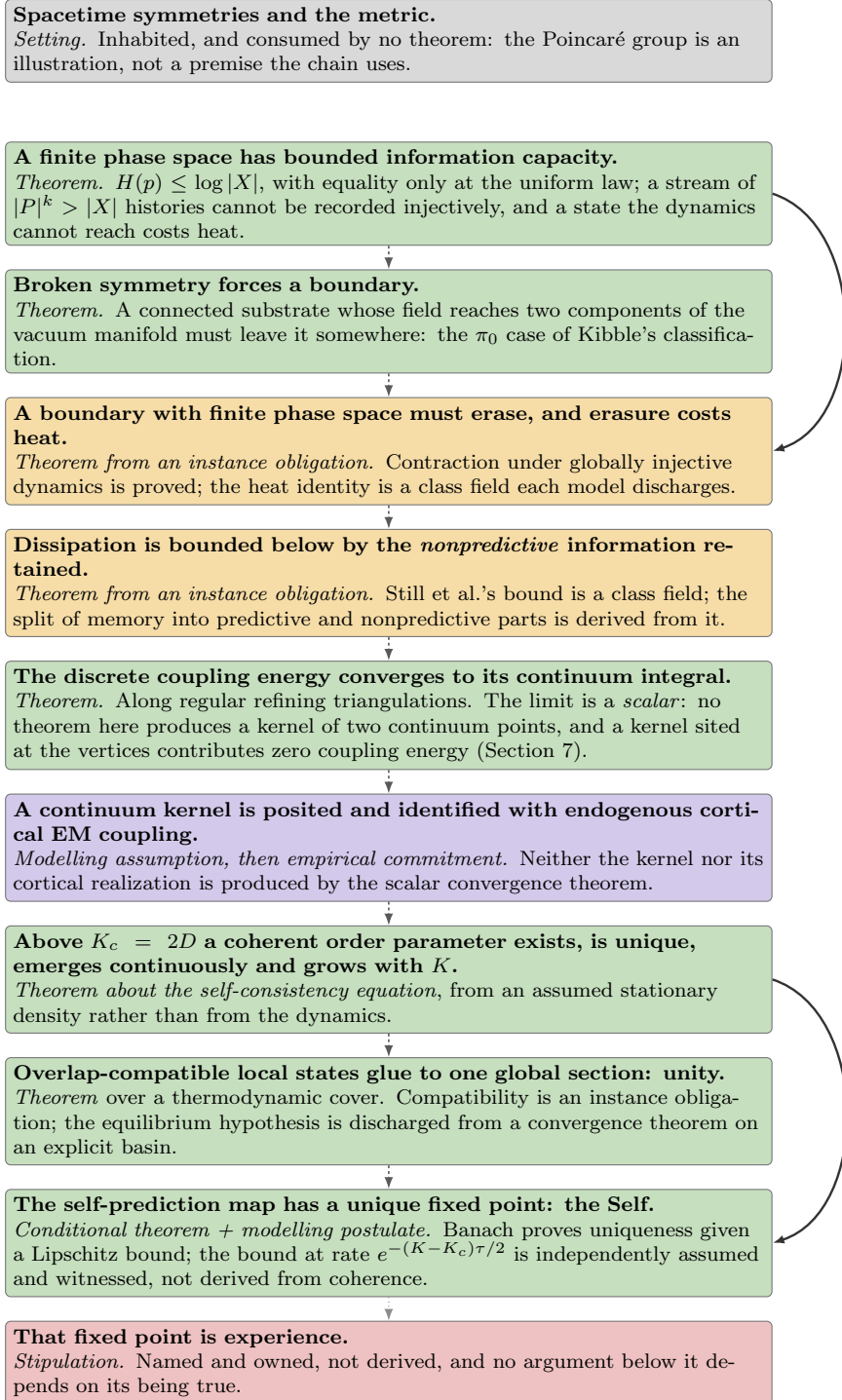


Figure 1: The conditional composition⁸ schema, with eight assumed vertical arrows and the status of each node and of each arrow. *Nodes.* ■ theorem; ■ theorem resting on a class field that each model must discharge; ■ empirical commitment, neither formalized nor derived; ■ stipulation; ■ setting, consumed by no theorem. *Arrows.* A **solid** arrow is a Lean theorem. A **dashed** arrow is a named hypothesis of chain (Chain.lean): an implication between the two node statements which the development does not prove

Claim	Status	What is assumed
A finite phase space has bounded information capacity	Theorem	Finiteness of the state space, and nothing else — no symmetry group, no metric. $H \leq \log X $, with equality only at the uniform law. On an infinite state space the step from “unreachable state” to “erasure” fails, and the failure is exhibited rather than assumed away.
Broken symmetry forces a boundary	Theorem	Substrate connected; vacuum manifold disconnected. No hypothesis mentions energy, a potential, or a symmetry.
Erasure costs heat	Theorem + class field	Global system–bath dynamics injective (Liouville). The heat identity <code>heat_eq</code> is discharged per instance, not asserted for all models.
Dissipation is bounded below by nonpredictive information	Theorem + class field	Still et al.’s bound is discharged per instance; the environment evolves by a Markov kernel the system does not act on.
The discrete coupling energy converges to its continuum integral; the field carrying that coupling is the cortical EM field	Theorem; then empirical	Convergence holds along regular refining triangulations, for the energy functional and not the dynamics. The limit is a scalar: no theorem produces a kernel of two continuum points, and a kernel sited at the triangulation’s vertices contributes zero coupling energy on an atomless substrate. The physical identification is a commitment with stated failure conditions (Section 7).
Above $K_c = 2D$ a unique coherent state exists, and overlap-compatible patches glue into a unique global section	Theorem 9	Von Mises stationary density assumed rather than derived from the Fokker–Planck operator; couplings positive and symmetric, so unfrustrated. Gluing needs a cover at the potential minimum whose synchronized patches agree on overlaps; the first is derived from a convergent trajectory, the second is an instance obligation throughout.
The coherent order parameter carries a unique reflexive fixed point:	Conditional theorem + modelling	The self-prediction map is the avatar’s read-out of its own encoding, and its Lipschitz rate

149 carry it, whose only hypothesis about the system is that its state space is
150 finite.

151 The first is the capacity bound. A probability distribution p on a fi-
152 nite state space X satisfies $H(p) \leq \log |X|$ (`shannon_entropy_le_log_card`),
153 with equality if and only if p is uniform (`shannon_entropy_eq_log_card_`
154 `iff`). The “only if” half is the part worth stating: $\log |X|$ is the exact capac-
155 ity rather than a convenient over-estimate. The proof is Gibbs’ inequality
156 against the uniform law, with the $p(x) = 0$ case carrying the slack that makes
157 the equality condition sharp.

158 The second says the limit binds. Let the boundary absorb a stream of
159 perturbations drawn from an alphabet P ; a stream of length k has $|P|^k$
160 possible histories, and once $|X| < |P|^k$ the map from histories to final states
161 cannot be injective (`absorb_not_injective`), so two distinct histories leave
162 the system in the same state (`source_entropy_exceeds_capacity`). The
163 update map is arbitrary and may be reversible at every individual step; the
164 collision is forced by counting alone, which is the sense in which the premise,
165 and not the model of the dynamics, does the work.

166 The third is what the thermodynamic step consumes. On a finite state
167 space, a map that fails to reach every state is not injective (`is_erasure_of_`
168 `not_surjective`) — which is exactly the erasure condition Landauer’s bound
169 requires — so any update of a finite phase space leaving some state unreach-
170 able dissipates strictly positive heat (`finite_phase_space_dissipates`). On
171 an infinite phase space the step simply fails: the successor map on $\mathbb{N}$ misses

0 and is injective. That failure is recorded beside the witnesses, so the finiteness hypothesis is fenced rather than assumed to be load-bearing. None of this derives the finiteness of phase space; finiteness is assumed, as in any statistical-mechanical treatment. What is removed is the gap between assuming the premise and using it.

4 Symmetry breaking and the inevitability of boundaries

A perfectly symmetric universe is the uniform vacuum; no distinct entities exist. For any dynamic process to occur, the continuous symmetries of the background field must undergo spontaneous symmetry breaking, and topology then dictates the creation of defects such as domain walls. This necessitates a physical boundary—a localized topological distinction between an “inside” and an “outside.” Such a defect is necessary to define a separate entity but not sufficient for agency: the boundary is relational, defined by the twisted geometry of the field surrounding it, and must be coupled with non-equilibrium thermodynamics to exhibit active behaviour.

That first claim is about topology, and it needs a different argument from the energy claim beside it. The energy half — an energy-minimizing field sits in the vacuum manifold almost everywhere (`pointwise_vacuum_of_global_min`) — together with the observation that a defect *obstructs* extension over a contractible interior, produces no defect: both are statements about a

193 field already given. What produces one is `exists_notMem_of_connected-`
 194 `ComponentIn`: if the substrate is connected and a continuous field takes val-
 195 ues in two *different connected components* of the vacuum manifold M , then
 196 somewhere the field is not in M at all. That excluded region is the wall.
 197 This is the π_0 case of Kibble’s classification (Kibble, 1976), and the proof
 198 is elementary in exactly the way it is elementary in the physics literature:
 199 the continuous image of a connected space is connected, and a connected
 200 subset of M lies inside one component of it. No hypothesis mentions energy,
 201 a potential, or a symmetry.

202 We formalize the π_0 case and stop there. The higher cases — $\pi_1 \neq 0$
 203 forcing vortex lines, $\pi_2 \neq 0$ forcing monopoles — are available in Mathlib’s
 204 homotopy library and not taken, because the argument this paper runs uses
 205 domain walls and nothing else. The witness is the double well, where the vac-
 206 uum manifold is *computed* to be exactly $\{-1, 1\}$ rather than merely known to
 207 contain both minima, and its disconnection is proved in the form the theorem
 208 consumes. `wellV_domain_wall` then quantifies over *every* continuous field on
 209 *any* connected substrate reaching both minima, with no formula for the field
 210 in the statement; that the class is non-empty is checked separately, since a
 211 theorem quantified over all fields is worthless if none satisfies its hypotheses.

212 5 The thermodynamic constraint

213 This topological boundary is subjected to gauge perturbations from the “out-
214 side” environment. Because the internal phase space is finite, it cannot ab-
215 sorb an unbounded sequence of incoming perturbations and must overwrite
216 old internal states. By Landauer’s principle (Landauer, 1961), the erasure of
217 physical information irreversibly decreases internal entropy, mandating that
218 the system exhaust heat into the environment.

219 Norton and Shenker have argued that *logical* erasure (many-to-one on
220 abstract computational states) need not dissipate heat if implemented by an
221 injective physical process (Norton, 2005; Shenker, 2000). Our formalization
222 addresses this by defining erasure as a non-injective mapping on the *physical*
223 state space and formalizing the global system–bath evolution as strictly in-
224 jective (Liouville). Heat dissipation is then a consequence of local state-space
225 contraction under globally reversible dynamics. The finiteness premise enters
226 exactly here: on a finite phase space, a state becoming unreachable is the
227 same as non-injectivity, and `finite_phase_space_dissipates` is the result.
228 The Norton–Shenker distinction is preserved rather than blurred: an update
229 that keeps the incoming perturbation on record is injective and dissipates
230 nothing, while one that overwrites it is not. Converting the combinatorial
231 statement to heat requires the class field `StatisticalMechanics.heat_eq`,
232 discharged by an explicit one-bit erasure system.

233 6 Prediction as the thermodynamic cost of 234 memory

235 A dissipative structure is driven by its environment. Write X_t for the state
236 of the structure and S_t for the environmental signal driving it, and let the
237 signal evolve by a dynamics of its own on which the structure does not act.
238 Two quantities are then defined at each step: the *memory* $I(X_t; S_t)$, and the
239 *predictive information* $I(X_t; S_{t+1})$, which is what that same state says about
240 the signal's next value.

241 **What the entropy-production bound does not give.** There is a nat-
242 ural route from entropy production to predictive processing, and it does not
243 work. Modelling the environment as a stochastic driving force $P(\text{ext})$ and
244 the internal transitions as $Q(\text{int})$, entropy production is bounded below by
245 $D_{KL}(P||Q)/\Delta t$ — this is Gibbs' inequality, carried in Lean as an instance
246 obligation. Rearranged it reads $D_{KL} \leq \Delta t \sigma$, so a system descending on σ
247 lowers a *bound* on the divergence, and it is tempting to conclude that the
248 internal transition probabilities come to mirror the environment's statistical
249 symmetries. They do not, because a driven system does not relax to $\sigma = 0$:
250 it relaxes to a non-equilibrium steady state at strictly positive entropy pro-
251 duction, and the bound then yields only $D_{KL} \leq \Delta t \sigma_{\text{NESS}}$, a positive number.
252 No limit may be read off it.

253 **Where predictive processing comes from.** Still, Sivak, Bell and Crooks
 254 (Still et al., 2012) prove that for a system driven by an environmental signal
 255 the work dissipated over one step of the drive obeys

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (1)$$

256 The bracket is the *nonpredictive* information: what the structure retains
 257 about the signal’s present that says nothing about its future. Memory
 258 that fails to predict is not merely useless to the structure; it is thermo-
 259 dynamic waste, quantified in energy. The bound is carried as the class field
 260 `PredictiveDissipation.still_bound`, for the same reason as every other
 261 physical postulate here: its `axiom` form, quantified over all joint laws with
 262 the dissipated work fixed, is refutable, and the development refutes it.

263 What is *derived* is everything downstream. First, the bracket is non-
 264 negative: $X_t \rightarrow S_t \rightarrow S_{t+1}$ is a Markov chain, so the data processing inequal-
 265 ity gives $I(X_t; S_{t+1}) \leq I(X_t; S_t)$ (`predictiveInfo_le_mutualInfo`). Second,
 266 dissipation is bounded below by a non-negative quantity, so $W_{\text{diss}} \geq 0$ follows
 267 rather than being assumed (`dissipatedWork_nonneg`). Third, the inference
 268 runs in the valid direction: $I(X_t; S_t) - I(X_t; S_{t+1}) \leq W_{\text{diss}}/k_B T$, so any bound
 269 on dissipation is immediately a bound on wasted memory (`nonpredictive_`
 270 `le_dissipation`). In the idealised quasi-static case every retained bit is
 271 predictive.

272 The no-back-action hypothesis is load-bearing and we exhibit what breaks

273 without it. In the formalization it is not an assumption stated beside the
 274 theorem but the shape of the evolution itself: the signal's step is the parallel
 275 kernel id $\parallel \kappa$, so no channel from X_t to S_{t+1} exists. A Markov kernel on the
 276 *pair* which copies the system's state into the signal carries an independent
 277 two-bit law to a perfectly correlated one, raising mutual information from
 278 zero to a positive value: an agent that steers its environment can know
 279 more about that environment's future than about its present, and Still's
 280 bound is not about such an agent. Both regimes are exhibited over one
 281 correlated two-bit joint law. With the signal frozen, every remembered bit
 282 is predictive and the instance may dissipate nothing, with the memory itself
 283 proved strictly positive; with the signal redrawn from its stationary law at
 284 each step, the entire memory is nonpredictive and the instance is *forced* to
 285 dissipate $k_B T$ times all of it. Kawai, Parrondo and Van den Broeck (Kawai
 286 et al., 2007) supply the corresponding equality — average dissipated work is
 287 exactly $k_B T$ times the relative entropy between forward and time-reversed
 288 phase-space densities — carried here as a predicate on an instance, which
 289 replaces the abstract dissipation field with a quantity an experiment can
 290 estimate (`nonpredictive_le_arrow`).

291 **What this does not establish.** It does not deliver the Free Energy Prin-
 292 ciple. Still's bound says that prediction is thermodynamically favoured; it
 293 says nothing about hierarchical generative models, variational free energy,
 294 or Bayesian inference. The Free Energy Principle (Friston, 2010; Friston &

295 Kiebel, 2009; Hohwy, 2013; Clark, 2016; Spisák & Friston, 2026) and the
 296 General Resonance Theory of Consciousness (Hunt & Schooler, 2019) are
 297 consonant with what is proved here, which is a weaker relation than deriva-
 298 tion and the only one we assert. Nor does it solve the symbol grounding
 299 problem (Harnad, 1990): maximal predictive information is compatible with
 300 the structure storing a lossy or unrecognisable function of the signal, and a
 301 state that predicts is not thereby a state that means. The narrower claim is
 302 the one worth making: whatever internal states a physical structure holds,
 303 those that do not help predict its environment are paid for in heat, so the
 304 pressure toward states indexed to the environment is energetic rather than
 305 semantic. We call the phenomenon *structural resonance*, and there is no
 306 teleological drive to “survive” in it: predicting the environment is simply the
 307 cheaper of the two ways to hold a memory.

308 **7 Macroscopic scaling: a theorem and a com-** 309 **mitment**

310 Three steps of different kinds meet here and are easy to run together: con-
 311 vergence of a scalar energy, which is a theorem; introduction of a continuum
 312 coupling kernel, which is a modelling assumption; and identification of that
 313 kernel with the brain’s endogenous electromagnetic field, which is an empir-
 314 ical claim about cortex. Neither later step inherits the licence of the first.
 315 Kept apart, the empirical claim is the framework’s central commitment and

316 the place where it is most easily wrong — which is where a claim of this kind
317 belongs.

318 **The formal half.** `Phase2_MeshConvergence.lean` proves that the discrete
319 coupling energy converges to a continuous integral along regular refining
320 triangulations (`mesh_refinement_convergence`). Because the summand is
321 symmetric, the double sum is the midpoint rule over unordered edges, so the
322 discrete energy is a quadrature of the continuous one. What this does not
323 establish: it is a statement about an energy functional, not about dynamics
324 — no trajectory convergence and no renormalization-group flow. The con-
325 tinuous object, `ContinuousNeuralField`, is *posited* with a coupling kernel
326 K , not produced by the theorem.

327 **The empirical step.** The formal half supplies scalar energy convergence
328 and no kernel. The field sections therefore posit a symmetric continuum
329 kernel as a modelling input. Which physical variable in cortex realizes that
330 independently supplied kernel is a further empirical question, and we state
331 our answer as a commitment:

332 *The model identifies its posited continuum kernel with the endoge-*
333 *nous electromagnetic field of mammalian cortex.*

334 This is a claim about biology, not a consequence of anything above it.
335 It is what makes the framework falsifiable and it is what we expect to be
336 attacked.

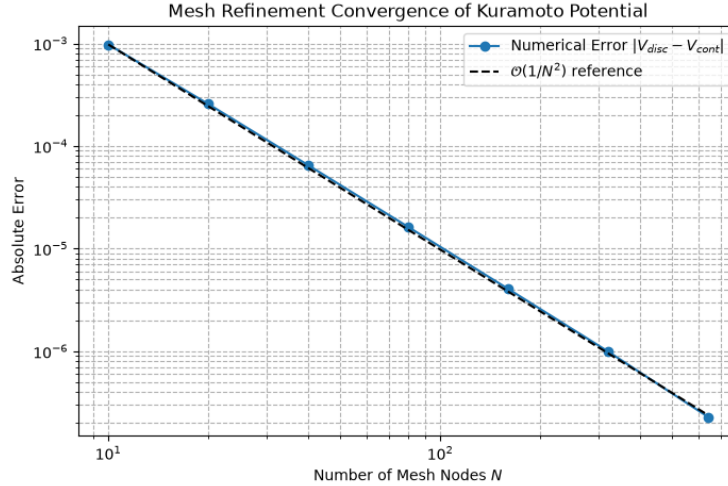


Figure 2: The coarse-graining theorem, exercised numerically. The discrete coupling energy of N units evenly spaced on the ring, with a Gaussian kernel of periodic distance and phase field $\theta(x) = \sin x$, is compared against the continuous integral it is proved to converge to. The error falls as $\mathcal{O}(N^{-2})$, the rate of the midpoint rule, which is the quadrature the Lean statement uses. The theorem asserts convergence, not a rate; the rate is what the simulation adds, and it is the ordinary rate for this quadrature rather than anything the framework predicts. The convergence is of an *energy functional* — nothing here says the discrete trajectories converge to the continuous ones. Parameters and scripts are in the supplement.

337 **What supports it.** Neurons do not only signal through synapses: trans-
338 membrane currents generate extracellular fields that shift membrane poten-
339 tials of neighbouring cells, a mechanism known as ephaptic coupling. En-
340 dogenous fields of 1–5 mV/mm entrain spike timing (Fröhlich & McCormick,
341 2010; Anastassiou & Koch, 2015), and stable electric fields carry information
342 that survives representational drift (Pinotsis & Miller, 2022). The field is
343 spatially continuous and superposes contributions from large populations —

344 a medium in which spatially separated activity is combined without further
345 mechanism. This is the integrative role the CEMI field theory assigns it
346 (McFadden, 2002, 2020; Pockett, 2012), physically supported by astrocytic
347 syncytia joined by gap junctions (Verkhratsky & Nedergaard, 2018).

348 **What the evidence supports, and what it does not.** The endogenous
349 field’s effect on any individual neuron is modulatory — it biases when a cell
350 fires; it does not decide whether it fires (Anastassiou & Koch, 2015). We
351 do not claim it is the *primary* dissipative structure. The claim is that it is
352 modulatory in effect and integrative in spatial reach: weak per synapse, and
353 the only variable defined everywhere at once. What the framework needs is
354 the reach, not the strength.

355 **Why reach is the property that matters.** A theory of consciousness
356 needs an account of why many local processes make one experience, and most
357 supply it by stipulation or by a measure. The account here is structural, and
358 it is the framework’s strongest argument. The field is the only variable in the
359 system that is *defined everywhere at once*. Unity is not asserted of it; unity
360 is what a field is. That is what makes the sheaf-theoretic step of Section 8
361 available at all — local states glue to one global section because there is one
362 object over which gluing can happen — so the answer to the combination
363 problem is a piece of geometry rather than a postulate. Compare “integration”
364 in Integrated Information Theory, which is a number, or “broadcast”
365 in Global Workspace Theory, which is a metaphor about a workspace. It is

366 also what the empirical identification is carrying: if the posited continuum
367 coupling is realized by something that is *not* defined everywhere at once —
368 by synaptic connectivity, say — then this argument is unavailable, and the
369 framework loses the property that distinguishes it from its rivals rather than
370 merely losing a mechanism.

371 **Its critics.** The longest-standing objection is that the fields are too weak
372 to matter, pressed from within the field-theoretic camp by Pockett herself
373 (Pockett, 2012, 2002). Vöröslakos and colleagues found that applied cur-
374 rent is shunted through scalp, delivering fields at or below 1 V/m in cortex
375 (Vöröslakos et al., 2018). That is a bound on *applied* rather than endogenous
376 fields, but it constrains how much causal work a field of that magnitude can
377 do, and is the reason we do not claim the field drives spiking.

378 **What would falsify it.** We state the failure conditions rather than leav-
379 ing them to a referee. **Epiphenomenality:** if the endogenous field can
380 be cancelled or shunted without altering transmembrane currents and nei-
381 ther behaviour nor report changes, the identification fails. **Magnitude gap:**
382 if the spatial integration required demands field effects an order of magni-
383 tude above the 1–5 mV/mm measured in cortex, the commitment fails on
384 arithmetic. **Timescale mismatch:** if recovery from sleep inertia tracks an
385 exponential rather than the delayed sigmoid the slow variable predicts, the
386 framework loses its own prediction. **Dissociation:** if large-scale field coher-
387 ence is preserved where unity is lost — or absent where it is retained — the

388 identification has no physical carrier.

389 **The regime the theorems cover.** The dissipative account — well-posedness,
390 the Lyapunov identity, monotone descent, and that the motion stops along
391 every trajectory — holds with no assumption on the sign of the coupling.
392 *Where* it stops does not, and the claims about the destination all live on
393 the positive side. The memory capacity of a frustrated cortical network is
394 therefore not supported by any theorem here, and we do not assert it as part
395 of the chain. The natural repair — letting local states agree on overlaps up
396 to a phase and asking whether the offset cocycle is a coboundary — fails
397 because locked, twisted and splay configurations all assign a phase to every
398 patch, so their offsets are automatically a coboundary.

399 Primate LFP phase maps contain phase singularities with integer winding
400 number, and human cortical spiral centres classify cognitive tasks (Townsend
401 et al., 2015; Xu et al., 2023). `Phase5.PhaseLifts` formalizes the measured
402 loop: transitions from one global real lift telescope to zero, while a three-
403 patch loop with winding one admits no global lift. The supplement gives the
404 measurement protocol and preregistered test this supports — a falsifiable
405 content test outside the conditional chain.

406 8 Global sections and the unity of conscious- 407 ness

408 The system is now characterized by a continuous macroscopic electrodynamic
409 field generated by many coupled micro-components, forced to resonate with
410 the environment. From here on the coupling regime is the unfrustrated one,
411 which is what every theorem below requires. The Kuramoto model (Ku-
412 ramoto, 1984) guarantees that past a critical coupling threshold these dis-
413 tributed local fields undergo a phase transition into unified synchronization
414 (Bauer et al., 2022), and `Phase4_KuramotoDynamics.lean` confirms that the
415 phase-locked state minimizes the Lyapunov potential — in the precise sense
416 that the functional minimized is the coupling potential read in the frame ro-
417 tating with the natural frequency, which `Phase4_RotatingFrame.lean` shows
418 is the honest reduced description of a system of identical oscillators. With
419 any natural frequency non-zero the full potential is unbounded below, so it
420 has no minimum for phase-locking to attain; the reduction is exact only for
421 identical frequencies, and a genuine spread leaves residual detunings whose
422 threshold is nowhere in our development.

423 The phase transition supplies synchronized phases. A separate compat-
424 ibility hypothesis says the corresponding local states agree on every over-
425 lap; given that, the sheaf condition glues them into a singular *global section*
426 (`Phase5_GlobalSection.lean`). The field acts as a single, indivisible topo-
427 logical space, providing the *unity of consciousness*—a necessary condition

428 for integrated experience. `ThermodynamicCover` demands that the cover's
 429 phase field minimize the reduced Kuramoto potential, and `potential_min_`
 430 `iff_phase_locked` characterizes those minimizers exactly, in both directions.
 431 Every instance is therefore already phase-locked, and the class's single con-
 432 dition on its local data — that patches at a common phase agree where
 433 they overlap — makes the family compatible. `probability_glue_unique`,
 434 the sheaf property of the probability presheaf, which contains no physics at
 435 all, then glues that family into exactly one global section.

436 The shape of the class makes this an emergence theorem rather than
 437 a uniqueness theorem. Were the class to carry a global section as a field,
 438 together with a second field declaring every local section a restriction of it,
 439 the global object would exist before the sheaf condition was consulted. It
 440 carries neither: the global section is constructed, and being a restriction is a
 441 *theorem*. The witness makes this concrete: two patches carry mass profiles
 442 $(2, 1, 7)$ and $(5, 1, 3)$ across three sites, agreeing only at the shared site. The
 443 theorem returns profile $(2, 1, 3)$ — neither of the two, and neither restricts
 444 correctly to both patches. The global state carries information no single
 445 patch had.

446 Two hypotheses carry physical content. That synchronized patches agree
 447 on their overlaps is a modelling assumption, not a consequence of any dy-
 448 namics we formalize. The other, `thermodynamic_equilibrium`, asks that the
 449 cover already sit at the potential minimum. That one is discharged rather
 450 than assumed, and the route is worth stating because it determines the scope

451 of the unity claim. `kuramoto_cauchy_problem` proves the Kuramoto initial
 452 value problem well posed on all of $\mathbb{R}$, so through every initial state there
 453 passes exactly one trajectory; along every trajectory of a zero-frequency sys-
 454 tem the dynamic potential converges and the total dissipation $\int_0^\infty \sum_i \dot{\theta}_i^2$ is
 455 finite; and since the dissipation rate is Lipschitz in time, Barbalat's lemma —
 456 which Mathlib does not carry, and which is proved here — gives $\sum_i \dot{\theta}_i^2 \rightarrow 0$,
 457 so the motion stops along every trajectory. *Where* it stops is taken condition-
 458 ally, by a route that avoids the LaSalle invariance principle, which Mathlib
 459 also does not carry. The flow is a gradient flow, so $\dot{V} = -\sum_i \dot{\theta}_i^2$ is an
 460 identity rather than an inequality; the minimizers of V are characterized ex-
 461 actly; and on the region where every pair of phases lies within a quarter turn
 462 a Polyak–Łojasiewicz estimate holds, $2a^2W \leq (\sum_{ij} A_{ij}) \sum_i \dot{\theta}_i^2$, with W the
 463 excess above the minimum and $a > 0$ a lower bound for the coupling. The in-
 464 variant region is produced by the Lyapunov function rather than assumed: a
 465 single pair contributes $\frac{1}{2}a(1 - \cos)$ to the excess, so an excess below $a/2$ forces
 466 every pairwise cosine positive for all time. Descent plus the estimate inte-
 467 grates to $W(t) \leq W(0)e^{-ct}$, which makes each $|\dot{\theta}_i|$ integrable on $[0, \infty)$, and
 468 a trajectory with integrable velocity converges. The conclusion (`kuramoto_`
 469 `tendsto_global_minimum`) is that if the phases start within a quarter turn
 470 of each other and the initial excess is below $a/2$, the trajectory converges to
 471 a phase-locked configuration which is a *global* minimizer, with order param-
 472 eter $r^2 = 1$. `ThermodynamicCover.ofConvergentTrajectory` then takes a
 473 cover whose phase field is the limit of such a trajectory and discharges the

474 equilibrium field *by that convergence*, with the cover’s coupling matrix the
475 coupling matrix of the dynamics rather than a separately chosen one. On
476 the resulting three-patch witness the starting configuration is provably not
477 phase-locked, so the configuration unity requires is arrived at rather than
478 posited.

479 The two hypotheses cannot be dropped, and this is the honest scope of
480 the result. Splay and twisted configurations are equilibria of the same flow,
481 so “every trajectory reaches the unified state” is false as a general statement,
482 and the initial-data condition is the standard arc condition under which
483 the true statement holds. What is claimed is conditional — unity is reached
484 from configurations already sufficiently coherent — and not that an arbitrary
485 initial state is carried into unity.

486 **Against IIT and GWT.** Our global section plays a structurally analogous
487 role to IIT’s Φ (Tononi & Koch, 2016) but is arrived at from the opposite end:
488 forced by a thermodynamic phase transition rather than postulated from phe-
489 nomenal axioms. The unity condition is checkable from field dynamics rather
490 than requiring the intractable computation of Φ , and exclusion falls out —
491 gluing either succeeds or it does not. Both are vulnerable to the objection
492 that the criterion could be satisfied by a system with no claim to experience
493 (Aaronson, 2014); our reply is that gluing is necessary but not sufficient, and
494 that the topological boundary and reflexive structure are additional required
495 conditions. Global Workspace Theory (Baars, 1997; Dehaene & Marti, 2014)

496 specifies a functional architecture without specifying what physical property
 497 makes broadcast content conscious (Block, 1995). Our framework supplies
 498 the missing substrate: the continuous field *is* the medium of integration, and
 499 the $K_c = 2D$ threshold gives a physical condition for ignition. Where GWT
 500 is agnostic about implementation, we make a falsifiable commitment — unity
 501 requires a continuous physical medium — which predicts that a functionally
 502 equivalent discrete broadcast architecture would not be conscious.

503 **9 Reflexive topology and the emergence of** 504 **the Self**

505 While a global section provides unity, a Self requires reflexivity. The unified
 506 field is a dissipative structure that must resonate with external perturbations
 507 while managing its own internal phase transitions, and time evolution drives
 508 it to minimize thermodynamic friction with its own state changes. This
 509 mandates *auto-resonance*: the field must predict itself. The Self emerges as a
 510 higher-order topological feature when the internal geometry folds to include a
 511 low-dimensional avatar of its own computational and causal boundary within
 512 its predictive model (`Phase6_ReflexiveTopology.lean`) (Levin, 2019).

513 **What carries the weight.** `reflexive_topology_implies_self` is the
 514 Banach fixed-point theorem with the physical commitments in its hypothe-
 515 ses: the space of global sections is assumed nonempty, metric and complete,

516 and the predictive map is assumed to contract. All four are discharged on
 517 the three-site substrate, and the metric is constructed rather than posited.
 518 The construction first pierces the sheafification: on a discrete substrate every
 519 point has a smallest open neighbourhood, so the mass a germ carries at a
 520 point is well defined, and `massEquivOn` proves that over any open the sections
 521 of the probability sheaf *are* the finite measures on that open, read as densities
 522 against counting measure. Mathlib supplies no inverse to this sheafification,
 523 so the correspondence had to be constructed. The metric on sections is then
 524 the uniform distance between densities, proved complete by exhibiting the
 525 limit section of a Cauchy sequence rather than by the degenerate route of
 526 showing Cauchy sequences eventually constant. The metric choice is load-
 527 bearing: under the 0/1 metric every contraction is constant (`contracting_`
 528 `implies_const`), and the Lévy–Prokhorov metric does not support this ar-
 529 gument on a discrete substrate, because the ε -thickening of a set is the set
 530 itself for $\varepsilon < 1$.

531 **The map is the avatar, or there is no map.** The predictive map is the
 532 composite of the avatar’s encoding and read-out, not free data alongside it.
 533 With the map free, Banach’s fixed-point theorem projects it out and never
 534 touches the avatar — replacing the avatar by a constant leaves the same Self.
 535 Three consequences follow from the composite structure. The Self lies in the
 536 range of the read-out (`self_mem_range_readout`), so it is reconstructed from
 537 a single section over the avatar region. Under the resonance condition $s =$

538 $\text{readout}(s|_{\text{avatar}})$: the field is what its own localized self-encoding reconstructs.
539 Blinding the avatar *moves* the Self: the blinded boundary's map is constant,
540 and the original Self is not a fixed point. If a family of resonant avatars *covers*
541 the substrate then the global state is determined by their local readings alone,
542 so unity is recovered from the avatars by the same sheaf condition.

543 **The contraction law is an independent modelling postulate.** The
544 rate $\exp(-(K - K_c)\tau/2)$ with $K_c = 2D$ is the linear relaxation rate of the
545 mean-field order parameter over an interval τ . The framework independently
546 assumes that the avatar-mediated self-prediction map is Lipschitz at this
547 same rate; no oscillator or field theorem derives that bound. Given the
548 postulate, the threshold results determine whether Banach can use it: above
549 threshold the rate is below one, so the assumed bound yields a unique fixed
550 point; at or below threshold the rate is at least one, and contraction at that
551 rate is false for every self-map of every metric space. This says only that
552 the Banach route is unavailable below threshold, not that a Self cannot exist
553 there. The postulate is witnessed at $K = 3$, $D = 1$, $\tau = 2 \ln 2$, but designing
554 a witness to realize the factor is not a derivation of self-model dynamics from
555 oscillator dynamics.

556 **The fixed point is the glued state.** The standalone Banach theorem
557 produces a fixed point somewhere in the space of global sections; by itself it
558 does not say that this is the section the sheaf condition glued. `Chain.lean`
559 preserves that identity. `IsUnifiedBy T s` says that s restricts to every local

560 state of the cover T , and the endpoint `UnifiedSelf` says that this same s
 561 is the unique fixed point of the reflexive map. The edge hypothesis must
 562 therefore supply the equality `predict s = s` as well as the Lipschitz bound;
 563 Banach supplies uniqueness but not the equality between the thermodynamic
 564 and reflexive constructions. On the witness, the two-patch cover glues ex-
 565 actly to the state that is already the unique fixed point of the non-constant
 566 avatar map. This exposes, rather than removes, the physical identification
 567 the composition requires. What is still not established is that any field dy-
 568 namics produces this particular read-out or performs the restriction, and that
 569 a cortex has the coupling the witness uses — the parameters are a choice of
 570 units, and nothing here derives a coupling constant from cortex.

571 **10 Continuous stochastic thermodynamics and** 572 **the field**

573 To move beyond finite graph analogies, we model the macroscopic electro-
 574 dynamic field as a *continuous neural field*, extending the discrete Kuramoto
 575 model to a continuous spatial coupling kernel $K(x, y)$ (`Phase8ContinuousField.lean`).
 576 Subject to thermodynamic noise D , the system undergoes non-equilibrium
 577 stochastic thermodynamics, and rather than a static Landauer bound the
 578 field dynamics minimize an entropy production functional

$$\sigma = \int \frac{1}{D} \left(\omega(x) + \int K_t(x, y) \sin(\theta_y - \theta_x) dy \right)^2 dx. \quad (2)$$

579 Biological fields dynamically deform their spatial topology $K_t(x, y)$ to un-
 580 dergo gradient descent on this functional, and that descent is a theorem
 581 rather than a modelling assertion. The drift is a fixed ω plus a linear operator
 582 applied to K , so σ is a quadratic functional of an affine image of the kernel. A
 583 squared norm is Fréchet differentiable on any real inner-product space, and an
 584 affine map is its own derivative, so the composite differentiates by the chain
 585 rule (`hasFDerivAt_quadratic_of_affine`). The drift map as a bounded op-
 586 erator $L^2(\mu \otimes \mu) \rightarrow L^2(\mu)$ exists by Cauchy–Schwarz, and its adjoint exists
 587 because L^2 is complete, giving $\nabla\sigma(K) = (2/D) A^*(AK + \omega)$. `structural_`
 588 `resonance_decreases_sigmaContinuum` is the descent result with no finite-
 589 ness hypothesis in it, run on Lebesgue measure restricted to $(0, 1)$, a substrate
 590 that is neither finite nor atomic. One limitation stays explicit: the phase field
 591 is held fixed throughout, so this is plasticity of the coupling at frozen phases,
 592 not joint (θ, K) dynamics.

593 A second result concerns where descent is headed, and is unconditional
 594 on kernel class. For a *reciprocal* kernel $K(x, y) = K(y, x)$, the integrand
 595 $K(x, y) \sin(\theta_y - \theta_x)$ is antisymmetric, so Fubini identifies its double integral
 596 with its own negative and the total coupling contribution vanishes. The total
 597 drift is therefore $\int \omega$ whatever the phases: coupling moves drift between sites
 598 but neither creates nor destroys it. Evaluating at the phase-locked config-
 599 uration pins its value, and the mean-matching hypothesis follows for every
 600 competitor (`phase_locked_minimizes_entropy_of_symm`). Reciprocity is the
 601 physically intended condition, but is not a formality: a *one-way* coupling on

602 two sites is strictly beaten, halving σ .

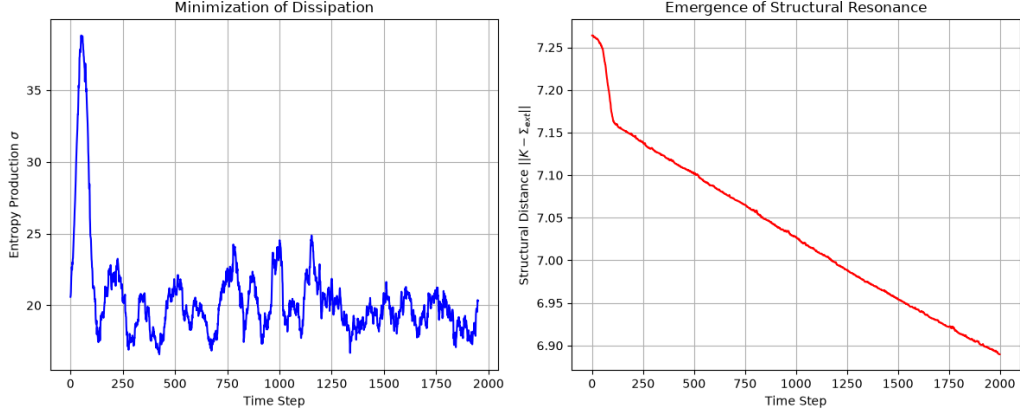


Figure 3: Coupling plasticity at frozen statistics: $N = 50$ units driven by perturbations drawn from a fixed environmental covariance Σ_{ext} , with the coupling matrix K descending the gradient of σ and clipped to stay non-negative and hollow. **Left:** entropy production, smoothed over 50 steps, falls from its transient and settles into a fluctuating plateau — a non-equilibrium steady state at strictly positive σ , not a relaxation to zero. That plateau is the reason this section declines the inference from $\sigma \rightarrow \min$ to $D_{KL} \rightarrow 0$: the system visibly does not reach $\sigma = 0$. **Right:** the distance $\|K - \Sigma_{\text{ext}}\|$ decreases monotonically over the run, the internal geometry drifting toward the statistics of the drive. Two cautions. The descent here is on σ at frozen phases, which is the regime the theorem covers; and the right panel falls by about 5% over 2000 steps, so what it shows is a direction of drift, not convergence. Parameters and scripts are in the supplement.

603 **The threshold.** The phase transition to the unified global section oc-
 604 curs when the mean-field spatial coupling exceeds the thermodynamic noise
 605 threshold $K_c = 2D$, linking the emergence of unity to physical temperature
 606 and noise. That predicate requires the substrate’s measure to be a prob-
 607 ability measure, and the hypothesis is load-bearing rather than decorative:

on a substrate of total mass m an unnormalized double integral carries a factor m^2 , so the comparison would track the substrate's size rather than its coupling, and a system a factor of two *below* threshold can clear the unnormalized bound. Under normalization, a constant kernel has mean-field strength exactly K and the predicate reduces to $K > 2D$.

The threshold $K_c = 2D$ is not a bare stipulation. The stationary density at order parameter r is von Mises $\rho(\theta) \propto e^{(Kr/D)\cos\theta}$, and self-consistency requires $r = R(K, r) := I_1(Kr/D)/I_0(Kr/D)$. Formalized as a ratio of integrals, integration by parts gives $\int \cos\theta e^{a\cos\theta} = a \int \sin^2\theta e^{a\cos\theta}$, so $R(a)$ is a times the mean of $\sin^2$ under the von Mises weight. The factor of two comes from $\mathbb{E}_a[\sin^2\theta] \leq 1/2$, proved by folding $[-\pi, \pi]$ onto $[0, \pi/4]$ where the integrand is a product of non-negative factors. At $K = K_c$ the map falls strictly below the diagonal for $r > 0$, so $r = 0$ is the only non-negative solution at threshold and the bifurcation happens strictly after K_c (`fixed_point_eq_zero_of_le_critical`).

Writing $R(a) = a \mathbb{E}_a[\sin^2\theta]$, continuity of $\mathbb{E}_a[\sin^2\theta]$ and $\mathbb{E}_0[\sin^2\theta] = 1/2$ give existence above $K_c = 2D$ by the intermediate value theorem (`supercritical_fixed_point_exists`). Uniqueness follows from $\mathbb{E}_a[\sin^2\theta]$ being strictly decreasing on $[0, \infty)$, proved without differentiation by folding both integrals onto $[0, \pi/2]$ and evaluating at a single crossing point. Together with continuity at threshold this gives the qualitative bifurcation diagram.

Differentiation is avoidable for uniqueness and unavoidable for the *rate*, and the rate is what Section 12 needs, so `Phase8_CriticalExponent.lean`

631 pays for it separately. Differentiating the cosine moments under the integral
 632 sign — dominated by the constant $e^{|a|+1}$ near the parameter — makes E twice
 633 continuously differentiable and evaluates $E(0) = 1/2$, $E'(0) = 0$ and $E''(0) =$
 634 $-1/8$ from elementary integrals of powers of $\cos$. Taylor's theorem gives
 635 $E(a) = \frac{1}{2} - a^2/16 + o(a^2)$, and substituting into the level-set equation gives
 636 the mean-field exponent: $r^2/(K - K_c) \rightarrow 1/D$ along any family of positive
 637 coherent solutions with $K \downarrow K_c$, so $\beta = 1/2$. Because those hypotheses hold
 638 only eventually along a filter, and an eventual clause is vacuously true on the
 639 empty one, the file discharges them on the branch itself rather than asserting
 640 that some family satisfies them.

641 One further step makes the equation mean in Lean what it means on
 642 paper. As a definition, `selfConsistency` is a function of r with a sugges-
 643 tive name. `vonMisesDensity` is proved a genuine probability density on
 644 $[-\pi, \pi]$; `circularOrderParameter` is the average of $e^{i\theta}$ against a density;
 645 that average is proved exactly $R(a)$, real because the density is even; and
 646 `fixedPoint_iff_selfReproducing` states the equivalence the name promises:
 647 r solves $r = R(K, r)$ if and only if the von Mises density it induces has order
 648 parameter r . Read through this, the existence theorem says that above K_c
 649 there is a stationary density whose own order parameter is positive.

650 **What remains not established.** The stationary density is an input, not
 651 derived from dynamics. Of the coherent branch we prove existence, unique-
 652 ness, continuous emergence at threshold and strict growth in K , but not that

653 it is dynamically selected. And the self-consistency equation is connected to
654 a *density*, not a *trajectory*: relating the two is propagation of chaos for the
655 finite Kuramoto system, a research programme rather than a lemma. That
656 single gap is what separates our threshold predicate from a statement about
657 dynamics.

658 Clinical evidence for this phase transition is observed during general anes-
659 thesia: propofol disrupts high-frequency rhythm phase alignment, breaking
660 structural resonance and forcing the integrated coupling below K_c (Bastos &
661 Miller, 2021).

662 11 Corollary: hardware thermodynamics and 663 topological rigidity

664 Both biological tissue and artificial silicon hardware are physical dissipative
665 structures. When a GPU computes, its charge carriers undergo physical
666 state transitions; by Landauer’s principle (Landauer, 1961), this erasure ex-
667 hausts heat into the environment. There is a thermodynamic distinction in
668 the degrees of freedom available to minimize this dissipation: in standard
669 von Neumann architectures the spatial topology is rigidly fixed, and the cou-
670 pling matrix A_{ij} is constrained by a discrete layout that cannot be physically
671 altered during operation.

672 The brain’s own architecture provides a suggestive constraint. The cere-
673 bellum contains approximately 80% of the brain’s neurons arranged in a dis-

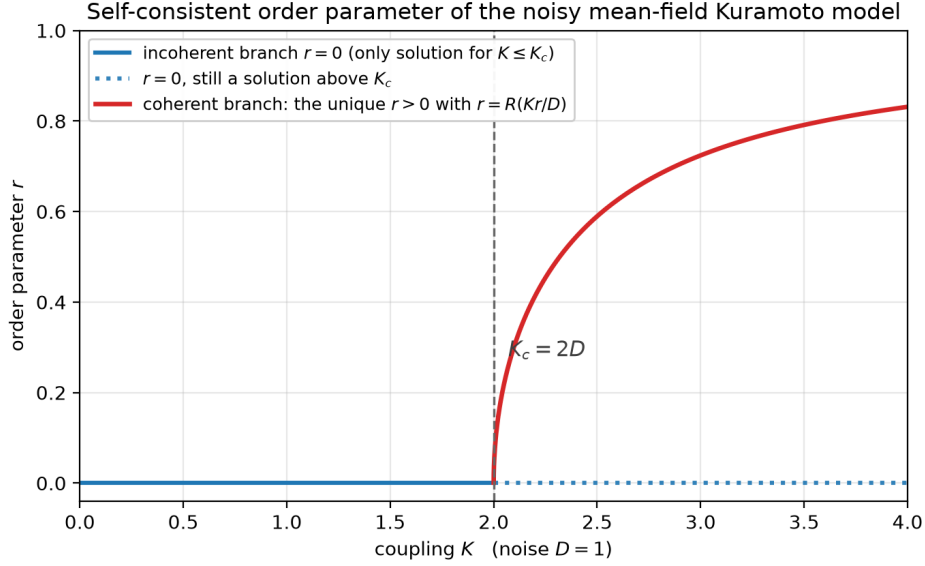


Figure 4: The self-consistency equation $r = R(Kr/D)$ of Section 10, drawn at $D = 1$. Every feature of this picture is a proved statement about the equation. At or below $K_c = 2D$ the incoherent state is the *only* non-negative solution, threshold included (`fixed_point_eq_zero_of_le_critical`); above it exactly one $r > 0$ joins it (`supercritical_fixed_point_existsUnique`), so the solution set is $\{0\}$ below and $\{0, r\}$ above; that branch tends to 0 as $K \downarrow K_c$ (`coherent_branch_continuous_at_threshold`), which is what makes the transition supercritical; and it increases strictly in K (`coherent_branch_strictMono`). The curve is computed by quadrature of the same integrals the Lean file defines, solving $E(a) = D/K$ for the concentration $a = Kr/D$ by bisection — legitimate precisely because E is proved strictly decreasing. It is an illustration of theorems, not evidence for them. What it does *not* show is stability: nothing in the development says the coherent branch is the dynamically selected one.

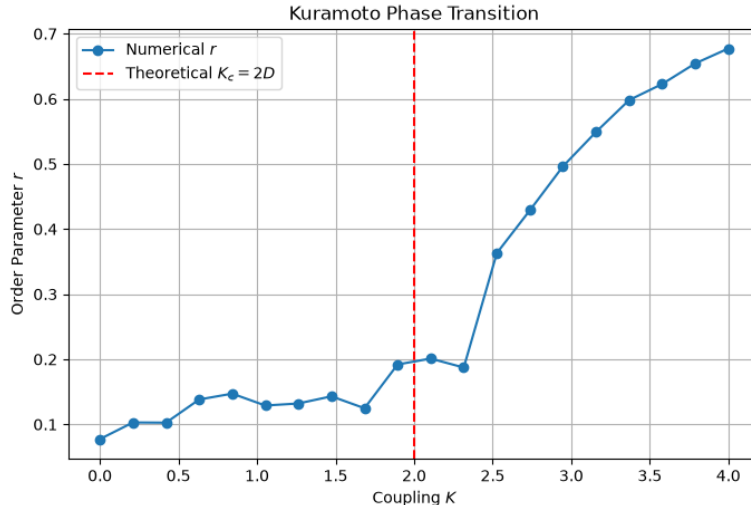


Figure 5: The same threshold reached by simulation rather than by analysis, and by a different route, which is why the two figures are shown side by side rather than merged. Here $N = 100$ oscillators with *no* noise carry natural frequencies drawn from a Lorentzian of scale $D = 1$, and the asymptotic order parameter is measured from the trajectories; the threshold for that ensemble is $K_c = 2D$ by Kuramoto’s original self-consistency argument. Figure 4 is the *noise*-driven model with identical frequencies, whose threshold is $2D$ by Sakaguchi’s (Sakaguchi, 1988), and it is the one the Lean file formalizes. The agreement of the two numbers is a known feature of the mean-field model, not a result of ours. The finite- N curve rises before the threshold and rounds off at it, as a finite sample must: $r \sim N^{-1/2}$ from incoherent phases alone.

674 crete, feedforward crystalline circuit (Herculano-Houzel, 2009; Lange, 1975),
675 and clinical observations of cerebellar agenesis report largely preserved con-
676 scious experience despite motor and cognitive deficits (Yu et al., 2015). This
677 dissociation has been read as evidence for the importance of recurrent archi-
678 tecture by IIT (Tononi & Koch, 2015) — an argument we borrow at the level
679 of the anatomical observation while declining its Φ -based explanation. We

680 take the dissociation as evidence, not as confirmation: it is a single anatom-
 681 ical case read against a theory, and it bears the weight such a case can bear.

682 Two formal results carry the corollary, and we state precisely what each
 683 answers. The first objection is that a reallocation lemma can assume its
 684 conclusion: a hypothesis such as “positive weight on a badly-correlated
 685 pair while a better one exists” is most of what one wants to prove about
 686 rigid hardware. It is not assumed here. `Phase7_Rigidity.lean` models
 687 an architecture by the set S of site-pairs it has a *wire* for and requires
 688 its couplings to vanish off S ; weights are tunable, wiring is not. Then
 689 `totalCorrelation_le_of_realizable` bounds every realizable coupling by
 690 $R \cdot M$, where R is the resource budget and M bounds phase correlation over
 691 the wired pairs; the bound is attained; and `rigid_is_strictly_suboptimal`
 692 *derives* the suboptimality hypothesis whenever the substrate’s best-correlated
 693 pair is one the wiring misses, with `rigid_gap` making it quantitative. This is
 694 resource-matched and not question-begging, but note what it turns on: fixed
 695 wiring support, not continuity. The second objection is that a development
 696 in which both architectures are matrices on one finite index type does not dis-
 697 tinguish a continuum from a lattice at all. On a substrate whose measure has
 698 no atoms, `fieldCorrelation_sited_eq_zero` shows a kernel supported on
 699 finitely many sites contributes *exactly zero* coupling energy, whatever weights
 700 it carries and whatever the phase field does, while a patch of strength c on U
 701 achieves $c\mu(U)^2$; on the real line, a thousand units of coupling concentrated
 702 at the origin register zero against a unit patch that registers one. What that

703 establishes is that the continuum coupling energy is blind to a finitely-sited
 704 architecture — a statement about which functional such an architecture can
 705 register in, not a demonstration that digital hardware computes worse.

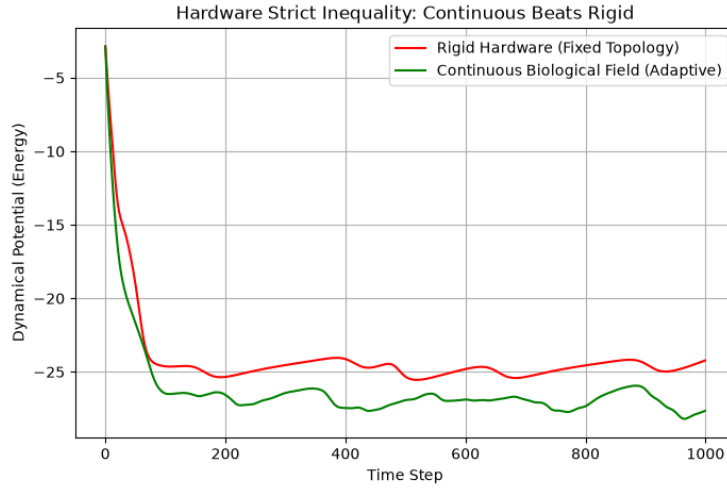


Figure 6: Rigid against deformable coupling at matched resource. Two copies of the same $N = 30$ oscillator system, same initial phases and same natural frequencies, differ only in whether the coupling matrix may move: the rigid copy keeps a fixed sparse routing matrix, the adaptive copy descends the gradient of the potential and is renormalized after every step so that $\sum_{ij} A_{ij}$ is held constant. The adaptive copy reaches a strictly lower potential and stays there. This is an illustration of the resource-matched suboptimality result and not a proof of it, and it is worth being clear about the gap: the theorem is about fixed wiring *support* — an architecture already wired to the best pair is not beaten — while the simulation shows a randomly wired one being beaten, which is the typical case rather than the general one. Parameters and scripts are in the supplement.

706 **What follows, and how far.** The step from either result to “von Neu-
 707 mann architectures cannot experience unified consciousness” remains an in-

708 formal argument, which we mark rather than blur. On our reading, a GPU’s
709 fixed routing matrix leaves it topologically fragmented, so the discrete layout
710 cannot supply the spatial coupling gluing into a global section requires, and
711 rigid hardware cannot undergo the reflexive auto-resonance that generates
712 a Self. This aligns with Searle’s biological naturalism: where the Chinese
713 Room argument asserted that discrete symbol manipulation cannot generate
714 understanding, our derivation supplies a candidate physical mechanism —
715 topological rather than semantic (Searle, 1980). The obstacle to pure sym-
716 bol manipulation is then physical rather than semantic: the substrate cannot
717 hold a phase-locked field.

718 The constraint does not preclude artificial consciousness; it restricts its
719 hardware. Analog field-based computing, neuromorphic architectures with
720 continuous memristive plasticity, and thermodynamic computing systems
721 (England, 2018; Ganesh & Parrondo, 2021) possess the spatial coupling re-
722 quired to physically morph their internal topology, which aligns with Hin-
723 ton’s concept of “mortal computation” (Hinton, 2022): architectures where
724 software is inextricably fused with the physics of the hardware. Biological
725 consciousness uses macroscopic continuous electromagnetic fields together
726 with a rich alphabet of molecular and transcriptomic cell states (Tasic et al.,
727 2018), which allows the coupling to deform across a continuous parameter
728 space.

729 12 The sleep-inertia timescale mismatch

730 Most of what this framework asserts is shared with some rival: that corti-
731 cal synchrony matters, that dissipation constrains cognition, that unity is a
732 physical rather than a computational property. One prediction is not shared,
733 and it follows from the framework’s specific commitment — that the medium
734 carrying the unified state is a continuous field whose effective coupling de-
735 pends on the physical geometry of the extracellular space, not on the rhythms
736 alone. Where that commitment differs from a purely electrical account, it
737 differs in an observable way, and we set out the difference precisely enough
738 to be wrong.

739 The daily loss of consciousness during sleep is modelled here as an active
740 physical disruption of the medium rather than a quiescence of the oscillators.
741 During sleep, astrocytic shrinkage expands the extracellular space, allowing
742 an influx of cerebrospinal fluid to flood the brain for metabolic waste clear-
743 ance (Xie & Nedergaard, 2013). This influx alters the spatial medium itself,
744 disrupting the continuous structural resonance and preventing the coherent
745 gluing that conscious unity requires on this account. The oscillators need
746 not stop for the field to fragment; the geometry they are coupled through is
747 what changes.

748 The prediction concerns sudden awakenings. When jolted awake, the lo-
749 cus coeruleus floods the cortex with noradrenaline. High-frequency electrical
750 rhythms attempt to phase-align in milliseconds (Bastos & Miller, 2021), but

751 the noradrenaline-driven astrocytic constriction required to squeeze the ex-
752 cess fluid out of the interstitial pathways operates on a timescale of seconds
753 to minutes (Xie & Nedergaard, 2013) — three to six orders of magnitude
754 slower. During that lag the electrical oscillators are synchronizing across a
755 waterlogged topology, and the fragmented structural resonance is what is ex-
756 perienceed as grogginess. The unified global section re-establishes only when
757 astrocytic volume finishes matching the state of the electrical rhythms.

758 **Why this implies a delayed sigmoid.** Write $K(t)$ for the effective cou-
759 pling, a monotone function of the reconstricting extracellular geometry, and
760 $r(t)$ for the order parameter. The electrical variable is fast: at every instant
761 the oscillators are at the configuration appropriate to the current K , so r
762 tracks the equilibrium branch $r(K)$. The slow variable $K(t)$ carries the time
763 course over 10^2 – 10^3 seconds. Section 10 establishes each piece as a theorem:
764 at or below K_c only the incoherent solution exists; above it the coherent so-
765 lution is unique, emerges *continuously* at K_c , and is strictly increasing in K .
766 Composing monotone $K(t)$ with a branch that is zero below K_c , rises contin-
767 uously from K_c , and saturates above yields three phases: a genuine *latency*
768 while $K(t) < K_c$; a continuous rise as the threshold is crossed; and satura-
769 tion. That is a delayed sigmoid, and the delay is not a fitting parameter —
770 it is the time the slow variable takes to reach K_c .

771 **What the alternative predicts.** An account on which grogginess is the
772 electrical realignment itself has no slow variable and no threshold. Its recov-

773 ery is exponential near the fixed point: monotone from $t = 0$, fastest at the
 774 start, with no flat foot. The discriminator is the presence or absence of an
 775 initial interval of no recovery, in data already collected.

776 **The foot has a square-root cusp.** A delayed sigmoid is a shape that any
 777 two-timescale account with a slow saturating gate can produce, so the com-
 778 parison above discriminates against less than it appears to. The bifurcation
 779 says more than that the coherent branch leaves zero continuously; it says
 780 at what rate. The second-order expansion $E(a) = \frac{1}{2} - a^2/16 + o(a^2)$, sub-
 781 stituted into the level-set form of the branch, gives the mean-field exponent
 782 in a path-independent form: along any family of positive coherent solutions
 783 with $K \downarrow K_c$ and $r \rightarrow 0$, $r^2/(K - K_c) \rightarrow 1/D$, that is $r \sim \sqrt{(K - K_c)/D}$,
 784 the pitchfork exponent $\beta = 1/2$. Composed with a coupling that crosses
 785 threshold at non-zero speed, so that $K(t) - K_c \approx \dot{K}(t_0)(t - t_0)$ just after
 786 the crossing time t_0 , the recovery leaves its foot as $r(t) \propto (t - t_0)^{1/2}$: a
 787 square-root cusp with *infinite initial slope*. An exponential has finite initial
 788 slope. So does a logistic. So does the composition of a smooth monotone
 789 gate with any branch that leaves zero linearly, which is what a generic sat-
 790 urating nonlinearity supplies. The cusp belongs to the bifurcation and not
 791 to the two-timescale structure, and it lives in the first instants after onset
 792 rather than in a fitted time constant — which makes it the sharpest available
 793 discriminator against a rival that reproduces the sigmoid by other means.

794 **A sharper form still, with no free parameters.** The stationary phase
795 density at order parameter r is von Mises at concentration $a = Kr/D$, and
796 the average of $e^{i\theta}$ under that density is exactly $R(a) = I_1(a)/I_0(a)$. Self-
797 consistency says the density reproduces its own order parameter, so at every
798 instant $r = I_1(a)/I_0(a)$, whatever $K(t)$ is doing. This yields three tests on a
799 single recording. *Shape*: the phase distribution is von Mises, testable against
800 the histogram. *Collapse*: plotting $a(t)$ against $r(t)$, the whole recovery must
801 fall on $r = I_1(a)/I_0(a)$ — the *same* curve for every subject and session, since
802 it contains no subject-specific quantity. Individual differences move a point
803 along the curve, not off it. *Threshold*: $a(t)/r(t) = K(t)/D$ with no calibra-
804 tion, so the measured ratio must exceed 2 wherever coherence is non-zero
805 and approach 2 at onset. The two estimates must be made independently: a
806 from regressing the log-histogram’s slope on $\cos \theta$, r from the resultant length
807 $|\langle e^{i\theta} \rangle|$.

808 **Falsifiability.** The parameter-free form tests the von Mises stationary den-
809 sity and the mean-field closure jointly. It does not test $K_c = 2D$ indepen-
810 dently of that ansatz: the crossing at $a/r = 2$ is a consequence of the same
811 density, so a failure of the collapse and of the threshold are one failure. The
812 prediction is falsified by: an exponential fit beating a delayed-sigmoid on
813 early recovery; an onset preferring exponent 1 over 1/2; a latency that does
814 not scale with extracellular volume; a phase distribution that is not von
815 Mises; or an (a, r) trace departing from $I_1(a)/I_0(a)$. The first two falsify

816 with no reference to the sleep-inertia setting at all.

817 **What this prediction assumes.** Four things. The identification of K
818 with a monotone function of extracellular geometry is a modelling assumption.
819 The threshold $K_c = 2D$ uses an assumed von Mises stationary density,
820 so the argument relies on the proved *existence and continuity* of the coherent
821 branch, not the numerical threshold. The square-root cusp needs the slow
822 variable to cross threshold at non-zero speed ($\dot{K}(t_0) \neq 0$), weaker than the
823 adiabatic separation. And the exponent theorem proves no stability or dynamical
824 selection — $r = 0$ remains a solution above threshold — so the time
825 course $r(t)$ is carried by the adiabatic assumption, not a trajectory theorem.
826 The prediction is about functional form, not a time constant: a flat foot of
827 definite shape whose width is set by the slow variable.

828 13 Conclusion

829 Human-like cognitive architectures do not require biological mysticism or
830 exotic sub-quantum physics (Hameroff & Penrose, 2014). While quantum
831 theories of consciousness often invoke delicate subatomic phenomena to explain
832 subjective unity, every component assembled here is classical: broken
833 symmetries, finite phase space, Landauer erasure, and gradient descent on entropy
834 production. What the formalization establishes is that the eight named
835 hypotheses of the conditional composition are jointly consistent and that the

836 framework's remaining physical commitments are localized and named —
837 not that classical fields are *sufficient* for consciousness or that the hypothe-
838 ses follow from finite capacity. Two links carry that weight, and not in the
839 same way. The unified state is proved to glue uniquely once a cover sits at
840 the minimum of the coupling potential; the motion is proved to stop along
841 every trajectory; and *where* it stops is proved conditionally, for initial data
842 within a quarter turn and below an explicit energy threshold, with a cover
843 built on that limit discharging the equilibrium hypothesis unity would oth-
844 erwise assume. What is not proved is the unconditional statement, and it
845 is false: splay and twisted configurations are equilibria of the same flow, so
846 the basin condition is the scope of the claim rather than slack in it. That a
847 physical cortex starts inside that basin is an assumption. The second link is
848 untouched by any of this: the identification of a global section of the prob-
849 ability sheaf with the unity of experience is a stipulation of the framework,
850 not a consequence of it.

851 Philosophically, this framework bridges the historical gap between type
852 identity theory (Smart, 1959) and computational functionalism (Putnam,
853 1967). The subjective mental state is identical to the physical thermo-
854 dynamic field, yet its qualitative nature is defined by the topological and
855 functional dynamics of structural resonance rather than by the specific bio-
856 chemistry of the brain. This constrains the *multiple realizability* of mind:
857 consciousness is not exclusively bound to biological carbon, but neither is
858 it substrate-neutral, because the substrate is what fixes which couplings are

859 physically available. What is proved is that fixed wiring is strictly suboptimal
860 at matched resource, and that a finitely-sited architecture registers exactly
861 zero coupling energy in the continuum functional; the step from either to a
862 claim about von Neumann architectures is an argument we make in prose
863 and do not formalize.

864 The framework bridges the historical gap between type identity theory
865 (Smart, 1959) and computational functionalism (Putnam, 1967): the sub-
866 jective mental state is identified with the physical thermodynamic field, yet
867 its qualitative nature is defined by structural resonance rather than specific
868 biochemistry. This constrains multiple realizability — consciousness is not
869 exclusively biological, but neither is it substrate-neutral. By locating the
870 phase transition at a threshold with closed-form value $K_c = 2D$, it aligns
871 with Seth’s “real problem” of consciousness (Seth, 2021): rather than bridg-
872 ing an unbridgeable metaphysical gap, it offers a topological architecture that
873 can explain, predict, and perturb the structural properties of experience.

874 **Code Availability**

875 The complete Lean 4 formalization, encompassing the derivation of all theo-
876 rems presented in this manuscript, as well as the accompanying Python sim-
877 ulations, are publicly available in the GitHub repository: [https://github.](https://github.com/bobaseb/physical_primitives_of_mind)
878 [com/bobaseb/physical_primitives_of_mind](https://github.com/bobaseb/physical_primitives_of_mind). Additional documentation
879 and web-accessible proofs are hosted at the project website: <https://bobaseb>.

880 `github.io/physical_primitives_of_mind/`. Each of the five computed
881 figures is produced by one script in `simulations/` and regenerates exactly;
882 Figure 1 is drawn rather than computed.

883 Contributions

884 The conceptualization, manuscript drafting, and Lean 4 formalizations were
885 heavily AI-assisted. The Google Antigravity CLI (Gemini 3.1, Claude 4.6
886 Sonnet) served as a collaborative reasoning partner. DeepSeek V4 Flash
887 and Claude Opus 5 (via OpenRouter/Nous Hermes Agent) assisted with
888 code generation, proof debugging, and structural analysis. Claude Code
889 carried out later formalization passes, figures, and manuscript restructuring.
890 Responsibility for every claim rests with the author.

891 References

- 892 Townsend, R. G., Solomon, S. S., Chen, S. C., Pietersen, A. N. J., Martin,
893 P. R., Solomon, S. G., & Gong, P. (2015). Emergence of complex wave
894 patterns in primate cerebral cortex. *Journal of Neuroscience*, 35(11), 4657–
895 4662.
- 896 Xu, Y., Long, X., Feng, J., & Gong, P. (2023). Interacting spiral wave pat-
897 terns underlie complex brain dynamics and are related to cognitive pro-
898 cessing. *Nature Human Behaviour*, 7(7), 1196–1215.

899 Pinotsis, D. A., & Miller, E. K. (2022). Beyond dimension reduction: Stable
900 electric fields emerge from and allow representational drift. *NeuroImage*,
901 253, 119058.

902 Bauer, L. G., et al. (2022). Quantification of Kuramoto Coupling Between
903 Intrinsic Brain Networks Applied to fMRI Data in Major Depressive Dis-
904 order. *Frontiers in Computational Neuroscience*, 16.

905 van Bree, S., et al. (2025). Processes and measurements: a framework for
906 understanding neural oscillations in field potentials. *Trends in Cognitive*
907 *Sciences*.

908 Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated informa-
909 tion theory: from consciousness to its physical substrate. *Nature Reviews*
910 *Neuroscience*, 17(7), 450-461.

911 Baars, B. J. (1997). In the Theater of Consciousness: The Workspace of the
912 Mind. *Oxford University Press*.

913 Dehaene, S., Charles, L., King, J. R., & Marti, S. (2014). Toward a compu-
914 tational theory of conscious processing. *Current Opinion in Neurobiology*,
915 25, 76-84.

916 Friston, K. (2010). The free-energy principle: a unified brain theory?. *Nature*
917 *Reviews Neuroscience*, 11(2), 127-138.

918 Kuramoto, Y. (1984). Chemical Oscillations, Waves, and Turbulence.
919 *Springer Series in Synergetics*, Vol. 19.

- 920 Scherf, M. (2026). A Formal Proof of Non-Duality: An Exposition of the
921 'OnlyOne' System. *Unpublished manuscript, PhilPeople*.
- 922 Inoué, T. (2026). On Brain as a Mathematical Manifold: Neural Manifolds,
923 Sheaf Semantics, and Leibnizian Harmony. *arXiv:2601.15320 [q-bio.NC]*.
- 924 Spisák, T., & Friston, K. (2026). Self-orthogonalizing attractor neural net-
925 works emerging from the free energy principle. *Neurocomputing*.
- 926 Pinotsis, D. A., & Miller, E. K. (2023). In vivo ephaptic coupling allows
927 memory network formation. *Cerebral Cortex*, 33(17), 9877-9890.
- 928 Pinotsis, D. A., Fridman, G., & Miller, E. K. (2023). Cytoelectric coupling:
929 Electric fields sculpt neural activity and "tune" the brain's infrastructure.
930 *Progress in Neurobiology*, 226, 102465.
- 931 Ruffini, G., Salvador, R., Tadayon, E., Sanchez-Todo, R., Pascual-Leone, A.,
932 & Santarnecchi, E. (2020). Realistic modeling of mesoscopic ephaptic cou-
933 pling in the human brain. *PLOS Computational Biology*, 16(6), e1007923.
- 934 Cunha, G. M., Corso, G., de Sousa, M. P. B., & Lima, G. Z. d. S. (2024).
935 Can ephapticity contribute to brain complexity?. *PLOS ONE*, 19(12),
936 e0310640.
- 937 Goldwyn, J. H., & Rinzel, J. (2016). Neuronal coupling by endogenous elec-
938 tric fields: Cable theory and applications to coincidence detector neurons
939 in the auditory brainstem. *Journal of Neurophysiology*, 115(4), 2033-2051.

- 940 McFadden, J. (2002). The conscious electromagnetic information (cemi) field
941 theory: the hard problem made easy?. *Journal of Consciousness Studies*,
942 9(8), 45-60.
- 943 Pockett, S. (2012). The electromagnetic field theory of consciousness: a
944 testable hypothesis about the characteristics of conscious as opposed to
945 non-conscious fields. *Journal of Consciousness Studies*, 19(11-12), 191-223.
- 946 Goff, P. (2019). *Galileo's Error: Foundations for a New Science of Con-*
947 *sciousness*. Pantheon Books.
- 948 Seth, A. (2021). *Being You: A New Science of Consciousness*. Penguin.
- 949 Seth, A. (2024). Conscious artificial intelligence and biological naturalism.
950 *PsyArXiv*.
- 951 Levin, M. (2019). The computational boundary of a “self”: developmental
952 bioelectricity drives multicellularity and scale-free cognition. *Frontiers in*
953 *Psychology*, 10, 2688.
- 954 Friston, K., & Kiebel, S. (2009). Predictive coding under the free-energy
955 principle. *Philosophical Transactions of the Royal Society B: Biological*
956 *Sciences*, 364(1521), 1211-1221.
- 957 Clark, A. (2016). *Surfing Uncertainty: Prediction, Action, and the Embodied*
958 *Mind*. Oxford University Press.
- 959 Hohwy, J. (2013). *The Predictive Mind*. Oxford University Press.

- 960 Prigogine, I., & Stengers, I. (1984). *Order Out of Chaos: Man's New Dialogue*
961 *with Nature*. Bantam Books.
- 962 McFadden, J. (2020). Integrating information in the brain's EM field: the
963 cemi field theory of consciousness. *Neuroscience of Consciousness*, 2020(1),
964 niaa016.
- 965 Fröhlich, F., & McCormick, D. A. (2010). Endogenous electric fields may
966 guide neocortical network activity. *Neuron*, 67(1), 129-143.
- 967 Anastassiou, C. A., & Koch, C. (2015). Ephaptic coupling to endogenous
968 electric field activity: why bother?. *Current Opinion in Neurobiology*, 31,
969 95-103.
- 970 Pockett, S. (2002). Difficulties with the electromagnetic field theory of con-
971 sciousness. *Journal of Consciousness Studies*, 9(4), 51-56.
- 972 Vöröslakos, M., et al. (2018). Direct effects of transcranial electric stimulation
973 on brain circuits in rats and humans. *Nature Communications*, 9, 483.
- 974 Tasic, B., et al. (2018). Shared and distinct transcriptomic cell types across
975 neocortical areas. *Nature*, 563(7729), 72-78.
- 976 Verkhratsky, A., & Nedergaard, M. (2018). Physiology of Astroglia. *Physio-*
977 *logical Reviews*, 98(1), 239-389.
- 978 Picard, M., & Shirihai, O. S. (2022). Mitochondrial signal transduction. *Cell*
979 *Metabolism*, 34(11), 1620-1638.

- 980 Harnad, S. (1990). The symbol grounding problem. *Physica D: Nonlinear*
981 *Phenomena*, 42(1-3), 335-346.
- 982 Tononi, G., & Koch, C. (2015). Consciousness: here, there and everywhere?
983 *Philosophical Transactions of the Royal Society B: Biological Sciences*,
984 370(1668), 20140167.
- 985 Aaronson, S. (2014). Why I Am Not An Integrated Information Theorist
986 (or, The Unconscious Expander). *Shtetl-Optimized*. Retrieved from [https:](https://scottaaronson.blog/?p=1799)
987 [//scottaaronson.blog/?p=1799](https://scottaaronson.blog/?p=1799).
- 988 Block, N. (1995). On a confusion about a function of consciousness. *Behav-*
989 *ioral and Brain Sciences*, 18(2), 227-247.
- 990 Herculano-Houzel, S. (2009). The human brain in numbers: a linearly scaled-
991 up primate brain. *Frontiers in Human Neuroscience*, 3, 31.
- 992 Lange, W. (1975). Cell number and cell density in the cerebellar cortex of
993 man and some other mammals. *Cell and Tissue Research*, 157(1), 115-124.
- 994 Yu, F., Jiang, Q. J., Sun, X. Y., & Zhang, R. W. (2015). A new case of
995 complete primary cerebellar agenesis: clinical and imaging findings in a
996 living patient. *Brain*, 138(6), e353.
- 997 Bastos, A. M., Donoghue, J. A., Brincat, S. L., Mahnke, M., Yanar, J., Cor-
998 rea, J., ... & Miller, E. K. (2021). Neural effects of propofol-induced uncon-
999 sciousness and its reversal using thalamic stimulation. *eLife*, 10, e60824.

1000 Hameroff, S., & Penrose, R. (2014). Consciousness in the universe: A review
1001 of the 'Orch OR' theory. *Physics of Life Reviews*, 11(1), 39-78.

1002 Hunt, T., & Schooler, J. W. (2019). The easy part of the hard problem: a
1003 resonance theory of consciousness. *Frontiers in Human Neuroscience*, 13,
1004 378.

1005 England, J. L. (2018). Dissipative adaptation in driven self-assembly. *Nature*
1006 *Nanotechnology*, 13(4), 287-295.

1007 Ganesh, N., & Parrondo, J. M. R. (2021). Thermodynamic computing with
1008 non-equilibrium systems. *Nature Physics*.

1009 Hinton, G. (2022). The Forward-Forward Algorithm: Some Preliminary In-
1010 vestigations. *arXiv preprint arXiv:2212.13345*.

1011 Searle, J. R. (1980). Minds, brains, and programs. *Behavioral and Brain*
1012 *Sciences*, 3(3), 417-424.

1013 Smart, J. J. C. (1959). Sensations and brain processes. *The Philosophical*
1014 *Review*, 68(2), 141-156.

1015 Putnam, H. (1967). Psychological predicates. In *Art, Mind, and Religion*.
1016 University of Pittsburgh Press.

1017 Xie, L., Kang, H., Xu, Q., Chen, M. J., Liao, Y., Thiyagarajan, M., ... &
1018 Nedergaard, M. (2013). Sleep drives metabolite clearance from the adult
1019 brain. *Science*, 342(6156), 373-377.

- 1020 Nagel, T. (1986). *The View from Nowhere*. Oxford University Press.
- 1021 Love, B. C. (2026). Consciousness, AI, and the Limits of Scientific Explana-
1022 tion. *arXiv preprint arXiv:2606.00226*.
- 1023 Norton, J. D. (2005). Eaters of the lotus: Landauer’s principle and the return
1024 of Maxwell’s demon. *Studies in History and Philosophy of Science Part B:*
1025 *Studies in History and Philosophy of Modern Physics*, 36(2), 375-411.
- 1026 Landauer, R. (1961). Irreversibility and heat generation in the computing
1027 process. *IBM Journal of Research and Development*, 5(3), 183-191.
- 1028 Kibble, T. W. B. (1976). Topology of cosmic domains and strings. *Journal*
1029 *of Physics A: Mathematical and General*, 9(8), 1387-1398.
- 1030 Sakaguchi, H. (1988). Cooperative phenomena in coupled oscillator systems
1031 under external fields. *Progress of Theoretical Physics*, 79(1), 39-46.
- 1032 Barbălat, I. (1959). Systèmes d’équations différentielles d’oscillations non
1033 linéaires. *Revue Roumaine de Mathématiques Pures et Appliquées*, 4, 267-
1034 270.
- 1035 Still, S., Sivak, D. A., Bell, A. J., & Crooks, G. E. (2012). Thermodynamics
1036 of prediction. *Physical Review Letters*, 109(12), 120604.
- 1037 Kawai, R., Parrondo, J. M. R., & Van den Broeck, C. (2007). Dissipation:
1038 The phase-space perspective. *Physical Review Letters*, 98(8), 080602.

- 1039 Banach, S. (1922). Sur les opérations dans les ensembles abstraits et leur ap-
1040 plication aux équations intégrales. *Fundamenta Mathematicae*, 3(1), 133-
1041 181.
- 1042 Coddington, E. A., & Levinson, N. (1955). Theory of Ordinary Differential
1043 Equations. *McGraw-Hill*, New York.
- 1044 Cover, T. M., & Thomas, J. A. (2006). Elements of Information Theory (2nd
1045 ed.). *Wiley-Interscience*, Hoboken, NJ.
- 1046 Shenker, O. R. (2000). Logic and entropy. *Philosophy of Science*, 67, S146-
1047 S152.